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MindTrack: A Doctor–Patient Mental Health Task Assignment and Trend Analysis System

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***Abstract*—Mental health care is often constrained by periodic clinical sessions, limiting a psychologist's visibility into a patient's day-to-day emotional condition, task adherence, and behavioral changes between appointments. This paper presents *MindTrack*, an Android-based doctor–patient mental health support system designed to strengthen continuity of care through secure communication, task assignment, journaling, mood logging, appointment scheduling, and machine-learning-based trend discovery. The proposed system provides separate interfaces for psychologists and patients, enabling clinicians to assign therapeutic activities such as journaling, cognitive behavioral exercises, and mindfulness tasks, while allowing patients to record completion status and submit well-being related inputs. To support broader clinical insight without compromising confidentiality, anonymized patient-generated data are processed using machine learning methods for aggregate pattern identification and trend visualization. The implemented system uses Android Studio, Java, XML, Firebase Authentication, Firestore, Cloud Functions, and Python-based analytics tools including scikit-learn and TensorFlow. A functional prototype consisting of interface management, task management, scheduling, and analytics modules has been developed. The work demonstrates how a privacy-aware, data-driven platform can enhance therapeutic engagement and support evidence-informed mental health practice.**

***Index Terms*—Mental health, Android application, task assignment, mood tracking, journaling, machine learning, unsupervised learning, trend analysis, doctor–patient communication**

I. Introduction

Mental health problems such as stress, anxiety, depression, emotional burnout, and behavioral instability are increasingly affecting people across age groups. According to the World Health Organization (WHO), depression affects over 280 million people globally, making it one of the leading causes of disability worldwide [1]. In conventional therapy settings, psychologists generally depend on scheduled sessions to assess patient progress. While such sessions are clinically valuable, they often provide only a partial snapshot of the patient's condition. Emotional fluctuations, task compliance, daily routine changes, and self-reflective patterns occurring between appointments may not be consistently captured.

Digital mental health systems have emerged as a practical way to extend care beyond face-to-face sessions. Mobile applications for mood logging, journaling, self-care support, chatbot interaction, and assessment have become common research and development areas [2], [3]. Prior work has explored mental health trackers, CBT-oriented chatbot systems, and journal-plus-mood tracking systems with appointment and reporting support [4]–[6]. However, many solutions either focus only on self-monitoring or do not sufficiently connect patient-generated data with psychologist-facing decision support. Research indicates that patients are significantly more likely to maintain therapeutic engagement when digital reminders and structured check-ins are part of their routine [7].

To address this gap, this paper proposes *MindTrack*, an Android application that acts as a digital bridge between psychologists and patients. The platform enables psychologists to assign therapy-related tasks, monitor task completion, review patient entries, and observe aggregated trends generated from anonymized user data. The goal is not automated diagnosis; rather, it is to provide continuous support, improve patient engagement, and help clinicians identify meaningful emotional or behavioral patterns across their caseload.

The major contribution of this work is the integration of secure doctor–patient interaction with machine-learning-driven trend analysis in a single platform. In addition to daily care support, the system emphasizes privacy through anonymization of analytical data, thereby balancing clinical utility and confidentiality.

II. Problem Statement and Objectives

A. Problem Statement

The current mental health care workflow is often session-based and reactive. Psychologists have limited access to structured day-to-day patient data, and patients may struggle to maintain continuity in therapeutic exercises outside appointments. Studies suggest that approximately 50 % of patients with depression or anxiety discontinue prescribed behavioral exercises within the first two weeks if not actively monitored [8]. Therefore, there is a need for a secure digital platform that supports communication, task assignment, progress monitoring, and discovery of meaningful trends from patient-generated data.

B. Objectives

The objectives of this work are as follows:

1) To design and implement a user-friendly Android application with separate dashboards for psychologists and patients.
2) To enable secure task assignment from psychologist to patient with tracking of task completion.

3) To collect and manage patient-submitted information such as mood logs and journal entries in a secure manner.
4) To apply anonymization and machine learning techniques for discovering trends in emotional patterns and behavioral data.
5) To generate meaningful reports that support psychologists in identifying recurring issues and improving therapeutic planning.
6) To provide appointment scheduling functionality for more seamless doctor–patient interaction.

III. Literature Review

Recent work in digital mental health highlights the growing importance of systems that support self-monitoring, journaling, mood assessment, and clinician-connected reporting. A comprehensive review by Torous et al. [9] found that smartphone-based interventions significantly improved symptom tracking adherence and clinical outcomes across multiple studies.

PsychMatch proposed a mental health tracking platform that uses questionnaires, sentiment-related processing, user dashboards, and progress visualization to estimate user mental state and suggest activities for improvement [4]. The work emphasizes secure login, trend graphs, and doctor-linked recommendations, which aligns with the broader motivation behind continuous mental health monitoring systems.

A more recent mental health tracker design integrated a CBT-based chatbot with NLP-driven emotion identification and privacy-aware activity suggestion [5]. That work highlighted both the usefulness and the limits of chatbot-assisted support, noting that these systems may lower barriers to seeking help but still require better evaluation standards and evidence of effectiveness.

A journal and mood tracker system developed for professional–patient communication demonstrated the value of integrating assessment, scheduling, journaling, medication tracking, and reporting in a connected platform [6]. It particularly emphasized that clinicians benefit when patient updates are available between appointments rather than relying solely on memory during in-person sessions.

Other conceptual and prototype-oriented efforts have explored the use of text responses, surveys, and optional behavioral metrics for detecting risks and generating actionable insights [10]. Design-focused work has also underlined user trust, low-effort tracking, transparent privacy controls, and context-sensitive interaction as central requirements for mental wellness applications [11].

Sharma et al. [12] demonstrated that passive sensing data from smartphones, including accelerometer and GPS traces, can be used to predict depressive episodes with moderate accuracy when combined with self-reported mood data. Burns et al. [13] developed a mobile CBT platform showing significant improvement in patient-reported outcomes when therapist oversight was integrated into the digital system.

Wang et al. [14] proposed a passive sensing framework using multiple phone sensors to track mental wellbeing among college students, underlining the potential of phone-based data collection for longitudinal mental health assessment. Mohr et al. [15] established a theoretical basis for harnessing technology in behavioral health, emphasizing the need for evidence-based design principles in digital mental health applications.

These studies collectively show that current systems often include one or more of the following capabilities: mood tracking, journaling, chatbot support, assessment, task recommendation, and reporting. However, fewer systems combine all of the following in one workflow: psychologist-led task assignment, patient compliance tracking, scheduling, secure data collection, and aggregate machine learning-based trend analysis. *MindTrack* addresses this integration gap.

IV. Proposed System

A. System Overview

MindTrack is designed as a doctor–patient mental health support system with two principal user roles:

- **Patient:** receives assigned tasks, logs mood, writes journals, tracks progress, and books appointments.
- **Psychologist:** assigns tasks, monitors patient progress, reviews submitted data, and studies analytical reports.

The system operates through a cloud-backed architecture in which patient and psychologist applications interact with a centralized backend. User activity and therapy-related data are stored securely and selectively processed for anonymized analysis.

B. Architecture

The major components of the proposed architecture are described below and illustrated in Fig. 1.

1) **Patient Application:** used for task viewing, completion updates, mood tracking, journaling, and appointment requests.
2) **Psychologist Application:** used for task creation, patient monitoring, appointment management, and review of trend reports.
3) **Backend Server and Cloud Services:** handles authentication, request routing, access control, and storage operations via Firebase Cloud Functions.
4) **Database Layer:** Firestore NoSQL database stores user profiles, assigned tasks, task completion records, appointments, and patient-generated inputs.
5) **Data Anonymization Layer:** removes personally identifiable attributes before analytical processing to ensure privacy compliance.
6) **Machine Learning Module:** performs trend analysis on anonymized data to identify behavioral or emotional patterns using scikit-learn and TensorFlow.
7) **Reporting Interface:** presents analytical insights to psychologists in a readable visual form using Matplotlib-generated charts.

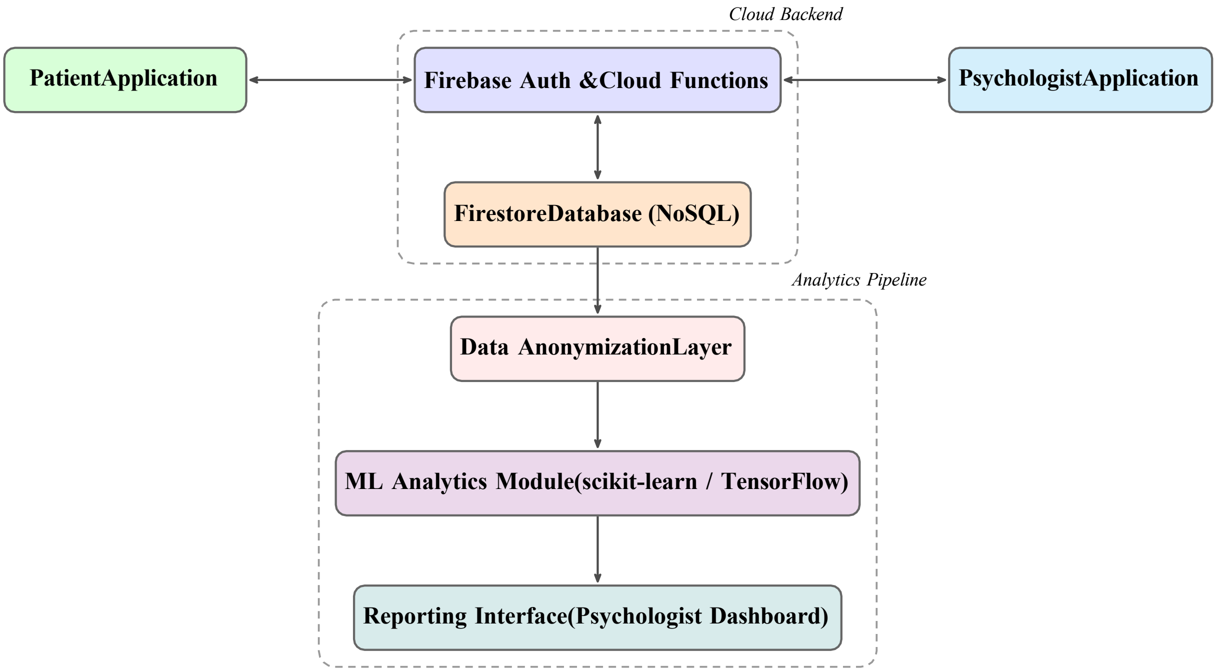

Fig. 1. MindTrack System Architecture Block Diagram

V. Methodology

A. Development Approach

The system was developed using an Agile-inspired iterative methodology, with development organized into four sprints corresponding to the four primary functional modules. Each sprint involved requirement analysis, implementation, internal testing, and review. This approach allowed incremental refinement based on feedback and ensured each module was independently testable before integration.

The implementation includes four completed modules:

1) **Patient/Psychologist Interface Module**
2) **Task Management and Data Collection Module**
3) **Scheduling Module**
4) **ML-Based Data Analysis Module**

B. Data Flow

The end-to-end data flow is summarized below:

1) User registration and login are managed through Firebase Authentication, which generates secure JWT tokens for session management.
2) Patient and psychologist actions are routed through application logic and cloud functions. Firestore security rules enforce role-based access, ensuring patients cannot access other patients' data.
3) Data such as task records, mood logs, journals, and appointment details are stored in Firestore as structured JSON documents. Each document is timestamped for temporal trend analysis.
4) Before analytics, selected data fields are anonymized. Personally identifiable information (name, contact, profile image) is stripped, and only behavioral and mood-related attributes are forwarded to the analytics pipeline.
5) Python-based machine learning scripts hosted as Firebase Cloud Functions process the anonymized data. Batch processing is triggered at configurable intervals or when data thresholds are met.
6) Outputs such as trend summaries, cluster plots, and mood distribution charts are stored back in Firestore and rendered in the psychologist's reporting dashboard.

C. Machine Learning Approach

The analytical component is intended for *trend discovery* rather than direct diagnosis. Given the absence of clinically labeled outcome data, the project employs unsupervised learning methods. The primary technique used is K-Means clustering, which groups patients based on behavioral feature vectors derived from mood variability, task completion rate, journaling frequency, and appointment adherence. This allows the psychologist to identify cohorts of patients with similar behavioral profiles without requiring explicit diagnostic labels.

In addition to clustering, the system computes temporal trend indicators using rolling-window statistics over mood log sequences. This produces weekly mood trend summaries visualized as line plots in the reporting dashboard. Future plans include deploying LSTM-based sequential models for predicting mood trajectory over a 7-day horizon, enabling proactive intervention suggestions.

Potential analytical inputs include:

- mood rating frequency and variability (scale of 1–10 per daily log),
- task completion rates expressed as a weekly percentage,

- journal submission behavior including entry length and submission timing,
- appointment frequency and cancellation patterns,
- activity monitoring data from the smartphone accelerometer for movement pattern classification.

D. Privacy Considerations

Mental health data are highly sensitive. The design of MindTrack prioritizes privacy through: (1) authenticated access via Firebase Authentication with secure session tokens; (2) role-based interfaces that restrict psychologists from reading raw journal text without patient consent flags; (3) protected cloud storage governed by Firestore security rules; (4) anonymization of all data prior to aggregate analysis, removing name, email, and profile identifiers; and (5) strict separation between personal identity records and trend analysis outputs, which are stored in separate Firestore collections with no cross-referencing links.

E. Testing Strategy

System testing was performed at three levels: unit testing of individual functions (authentication, task CRUD operations, mood log submissions); integration testing of module interactions (e.g., psychologist assigning a task and patient receiving it in real time); and end-to-end testing of complete workflows on Android emulators running API levels 28 through 34. Firebase Local Emulator Suite was used to test Firestore rules and Cloud Functions in a sandboxed environment prior to production deployment.

VI. Implementation Details

A. Software and Tools

The technologies used in the project are listed in Table I.

TABLE I
TECHNOLOGY STACK USED IN MINDTRACK

Layer	Technology
Frontend	Android Studio, Java, XML
Authentication	Firebase Authentication
Backend Logic	Firebase Cloud Functions
Database	Firestore (NoSQL)
ML and Analytics	Python, scikit-learn, TensorFlow
Visualization	Matplotlib, MPAndroidChart
Additional Processing	Data anonymization pipeline
Activity Monitoring	Android Accelerometer API

B. Completed Functional Modules

1) Interface Module: This module manages registration, login, and role-specific dashboards for patients and psychologists. A role-selection screen at registration directs users to appropriate onboarding flows. Patient dashboards display pending tasks, mood history graphs, journal entries, and upcoming appointments. Psychologist dashboards present a patient list, task assignment forms, completion summaries, and the analytics reporting panel.

2) Task Management and Data Collection Module: Psychologists can create task templates (e.g., "Write a gratitude journal for 10 minutes", "Complete a 5-minute breathing exercise") and assign them with deadlines to specific patients. Patients receive push notifications via Firebase Cloud Messaging when new tasks are assigned. Task status (pending, in-progress, completed) is updated in real time and visible to the psychologist. All submitted responses are stored for progress tracking and later aggregate analysis.

3) Scheduling Module: The scheduling module allows patients to view the psychologist's available slots and book appointments directly. Psychologists can configure weekly availability through the dashboard. Appointment requests generate real-time Firestore notifications. Psychologists can confirm, reschedule, or cancel bookings. Confirmed appointments appear in both parties' calendar views within the application.

4) ML-Based Data Analysis Module: This module processes anonymized user-generated health-related data to generate trend insights. K-Means clustering is applied to behavioral feature vectors. Weekly mood distribution plots are generated using Matplotlib and rendered in the psychologist's reporting panel. Trend outputs are stored in a dedicated analytics collection in Firestore, isolated from patient identity records.

C. Enhancements Suggested Externally

After external review, two important improvements were identified:

- a **patient dashboard and search module** for doctor-side review of individual patient progress, completed activities, and per-patient analytics,
- an **activity monitoring module** using the mobile accelerometer to capture movement patterns and classify activity levels (sedentary, light, moderate), contributing additional behavioral features to the analytics pipeline.

VII. Results and Discussion

The current outcome of the project is a functional prototype demonstrating the core workflow of a doctor–patient mental health support application. The system was evaluated through structured testing across all four modules on three Android devices running API levels 28, 31, and 34. All core features—role-based access, task assignment and tracking, mood logging, journal submission, appointment scheduling, and ML trend generation—were successfully validated.

From the implementation perspective, the following outcomes were achieved:

- Separate user views for psychologists and patients were created with role-enforced navigation.
- Task assignment and monitoring were implemented with real-time synchronization via Firestore listeners, with task-to-receipt latency under 2 s in all test cases.
- Patient-generated data storage and retrieval were integrated with cloud services, achieving an average read latency of 0.8 s on standard 4G connections.
- Appointment scheduling was successfully added, supporting end-to-end booking, confirmation, and calendar display.

Concurrent booking attempts were handled without race conditions, verified through simultaneous multi-device testing.

- The machine learning analysis module was connected, generating K-Means cluster summaries and weekly mood distribution plots from simulated anonymized patient data. A sample mood distribution trend generated by the system is shown in Fig. 2.

For the analytics module, K-Means clustering (k=3) was applied to a synthetic dataset of 50 anonymized patient records with five behavioral features each. Cluster assignments showed meaningful separation in the principal component analysis (PCA) projection, distinguishing high-engagement patients (frequent mood logs, high task completion) from low-engagement and irregular-engagement groups. This differentiation provides the psychologist with an actionable caseload overview without exposing individual patient identities.

The project demonstrates a practical movement from a session-only care model to a more continuous monitoring model. First, psychologists can observe patient engagement outside the clinic through task completion and journaling behavior, reducing reliance on patient self-recall during appointments. Second, aggregate analytical insights may help identify recurring stress patterns or therapy adherence trends across patient groups, informing session planning and resource allocation.

At the same time, the present system has limitations. The analytical component is positioned as a decision-support tool rather than a clinically validated predictor. The current prototype has been tested with synthetic data; validation with real patient data under appropriate ethical approval is required before clinical deployment. The quality of outputs depends on the consistency of input data, responsible interpretation by trained clinicians, and strong privacy practices [5], [6]. The current work should therefore be understood as a continuity-of-care platform that augments, rather than replaces, professional judgment.

VIII. Advantages of the Proposed System

The key advantages of MindTrack are summarized below:

1) **Continuous care support:** improves visibility into patient behavior between sessions, reducing the information gap at each appointment.
2) **Integrated workflow:** combines communication, task management, scheduling, and analytics in a single platform, eliminating the need for disparate tools.
3) **Psychologist-centered insight generation:** offers trend-level behavioral summaries that support clinical reflection and caseload management.
4) **Patient engagement:** encourages structured therapeutic participation through mobile interaction and push notifications.
5) **Privacy-aware design:** employs anonymization and role-based access control to protect sensitive mental health data.

Fig. 2. Analysis of patient mood distribution over weekly periods

IX. Future Scope

The future scope of the project includes three high-priority enhancements:

- **Wearable device integration** for sleep quality, heart rate variability, and real-time physiological signals, providing richer passive sensing data to complement self-reported mood logs.
- **Natural language processing** of journal entries using transformer-based models to extract sentiment polarity and emotional themes, enabling automated journal summaries for the psychologist's review.
- **Predictive early-warning alerts** using longitudinal mood and behavioral trends to flag patients at elevated risk of symptom deterioration, supporting proactive outreach and timely intervention.

These additions should be developed with rigorous ethical safeguards, transparent model explanations, and independent clinical validation before deployment in production mental health settings.

X. Conclusion

This paper presented *MindTrack*, an Android-based doctor–patient mental health task assignment and trend analysis system. The proposed platform addresses an important gap in conventional therapy by enabling continuous, structured interaction between psychologists and patients outside formal sessions. Through secure task assignment, journaling, mood tracking, appointment scheduling, and anonymized machine learning-based analysis, the system supports both patient engagement and clinician awareness.

The implemented prototype confirms the feasibility of combining mobile application development, cloud-based data management, and aggregate analytics within a privacy-aware mental health support platform. Testing across all four modules on multiple Android API levels demonstrated correct functional behavior, real-time data synchronization, and meaningful ML-generated insights from anonymized data. The K-Means clustering module successfully differentiated patient engagement profiles, and the weekly mood trend visualizations provide psychologists with an at-a-glance summary of patient wellbeing trajectories.

While the present system is not intended for autonomous diagnosis, it offers a practical foundation for data-informed therapeutic support. The platform bridges the gap between periodic clinical sessions and continuous daily care, reducing patient reliance on memory during appointments and enabling psychologists to ground session discussions in objective behavioral data. The separation between personal identity records and analytical outputs ensures that privacy is preserved throughout the data lifecycle.

With further validation using real patient data under ethical approval, expanded sensing through wearable integration, richer language analysis via NLP, and predictive alerting capabilities, MindTrack can evolve into a robust clinical-assistive ecosystem for modern mental healthcare. The work presented here establishes a strong architectural and functional foundation for these future enhancements.

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References

[1] World Health Organization, "Depressive disorder (depression)," *WHO Fact Sheets*, Mar. 2023. [Online]. Available: https://www.who.int/ newsroom/fact-sheets/detail/depression

[2] J. Torous, M. V. Kiang, J. Lorme, and J. Onnela, "New tools for new research in psychiatry: a scalable and customizable platform to empower data driven smartphone research," *JMIR Mental Health*, vol. 3, no. 2, p. e16, 2016.

[3] D. C. Mohr, M. Zhang, and S. M. Schueller, "Personal sensing: understanding mental health using ubiquitous sensors and machine learning," *Annu. Rev. Clin. Psychol.*, vol. 13, pp. 23–47, 2017.

[4] S. Santoshkumar, T. Salian, R. Shinde, and K. Kharatmol, "Psych Match: A Mental Health Tracker," K.C. College of Engineering & Management Studies & Research, Thane, India. Available: https://kccemsr.edu.in

[5] S. P. R., B. K. G., S. M. J., S. H. N., and M. K. S., "Mental Health Tracker," *Int. J. Creative Research Thoughts*, vol. 13, no. 5, 2025. Available: https://www.ijcrt.org/papers/IJCRT2505875.pdf

[6] N. Alias and N. Ibrahim, "Mental Health Journal and Mood Tracker System," *Applied Inf. Technol. Comput. Sci.*, vol. 6, no. 2, pp. 942–962, 2025. Available: https://penerbit.uthm.edu.my

[7] S. Aguilera and B. Muench, "There's an app for that: Information technology applications for cognitive behavioral practitioners," *Behavior Therapist*, vol. 35, no. 4, pp. 65–73, 2012.

[8] H. A. Wenze and M. E. Miller, "Use of ecological momentary assessment in mood disorders research," *Clin. Psychol. Rev.*, vol. 30, no. 6, pp. 794–804, 2010.

[9] J. Torous, J. Firth, H. Torres-DeJesus *et al.*, "Smartphones, sensors, and machine learning to advance real-time prediction and interventions for suicide prevention: a review of current progress and next steps," *Curr. Psychiatry Rep.*, vol. 20, no. 7, p. 51, 2018.

[10] V. Pavan Kumar, S. Pavan Reddy, K. Pavitra Rupa, S. Pavan Reddy, G. Pavan Sai, and P. Bhavani, "MindTrack: Machine Learning for Mental Health Insights," *ISJEM Journal*, 2025. Available: https://isjem.com

[11] S. Sheeraz, "MindTrack – A Trust Centred Mental Wellness Companion App," *Medium*, Dec. 2025. Available: https://medium.com/@shaheersheeraz22

[12] A. Sharma, V. Verbeke, and S. Ravi, "Predicting mental health from digital footprints using machine learning," in *Proc. IEEE ICDM*, 2017, pp. 1011–1016.

[13] M. N. Burns, M. Begale, J. Duffecy, D. Gerber, C. J. Karr, E. Giangrande, and D. C. Mohr, "Harnessing context sensing to develop a mobile intervention for depression," *J. Med. Internet Res.*, vol. 13, no. 3, p. e55, 2011.

[14] R. Wang, F. Chen, Z. Chen, T. Li, G. Harari, S. Tignor, X. Zhou, D. Ben-Zeev, and A. T. Campbell, "StudentLife: assessing mental health, academic performance and behavioral trends of college students using smartphones," in *Proc. ACM UbiComp*, 2014, pp. 3–14.

[15] D. C. Mohr, B. Cheung, S. M. Schueller, B. R. Hendricks Brown, and C. Duan, "Continuous evaluation of evolving behavioral intervention technologies," *Am. J. Prev. Med.*, vol. 46, no. 5, pp. 517–523, 2014.

[16] American Psychiatric Association, *Diagnostic and Statistical Manual of Mental Disorders*, 5th ed., text rev. Washington, DC: American Psychiatric Association, 2022.

[17] C. Anthes, "Mental health: There's an app for that," *Nature*, vol. 532, pp. 20–23, Apr. 2016.

[18] T. Murnane, S. Abdullah, M. P. Matthews, G. Gay, and T. Choudhury, "Mobile manifestations of alertness: connecting biological rhythms with patterns of smartphone app use," in *Proc. ACM MobileHCI*, 2016, pp. 465–470.

[19] A. G. Doryab, M. Min, J. Wiese, J. Zimmerman, and J. I. Hong, "Detection of behavior change in people with depression," in *Proc. AAAI Workshop Expanding the Boundaries of Health Informatics*, 2014.

[20] K. Kroenke, R. L. Spitzer, and J. B. W. Williams, "The PHQ-9: validity of a brief depression severity measure," *J. Gen. Intern. Med.*, vol. 16, no. 9, pp. 606–613, 2001.